Sampling & Estimation

Section 1: Unbiased Estimators of Population Mean & Variance

1 What is the unbiased estimator of the population variance σ^2 calculated from a random sample of size n?

a. $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$

b. $s^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum(x_i - \bar{x})^2$

d. $s^2 = \sum(x_i - \bar{x})^2$

2 A random sample of 10 observations has $\sum x = 80$ and $\sum x^2 = 724$. Calculate the unbiased estimate of the population variance.

a. 9.33

b. 8.40

c. 10.50

d. 7.20

3 If the biased sample variance is $v = 18$ from a sample of size $n = 25$, find the unbiased estimate s^2.

a. 18.75

b. 17.28

c. 18.00

d. 19.50

Section 2: The Central Limit Theorem (CLT)

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4 State the Central Limit Theorem for the distribution of the sample mean $\bar{X}$.

a. For large n, $\bar{X}$ is approximately normal $N(\mu, \sigma^2/n)$ regardless of the population distribution

b. For any n, $\bar{X}$ is strictly normal

c. For large n, the population becomes normal

d. $\bar{X}$ has mean $n\mu$ and variance $n\sigma^2$

5 A population has mean 100 and standard deviation 15. A sample of size $n = 36$ is taken. What is the standard error of the sample mean?

a. 2.5

b. 15

c. 4.17

d. 0.417

6 Using the distribution from the previous question ($\mu = 100$, SE $= 2.5$), find $P(\bar{X} > 105)$.

a. 0.0228

b. 0.0500

c. 0.0456

d. 0.0100

Section 3: Confidence Intervals for the Population Mean

7 A sample of size $n = 64$ from a normal population with known $\sigma = 8$ has sample mean $\bar{x} = 52.0$. Find the 95% confidence interval for μ.

a. $[50.04, 53.96]$

b. $[49.50, 54.50]$

c. $[50.36, 53.64]$

d. $[51.00, 53.00]$

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8 What is the critical value z used to construct a 99% confidence interval for the population mean?

a. 2.576

b. 1.960

c. 1.645

d. 2.326

9 How large must a sample be so that the width of a 95% confidence interval for μ is at most 2 units, given $\sigma = 10$?

a. At least 385

b. At least 100

c. At least 196

d. At least 250

Section 4: Confidence Intervals for Large Unknown-Variance Samples

10 When σ^2 is unknown but the sample size $n = 100$ is large, what is substituted for σ^2 in calculating the confidence interval?

a. The unbiased sample variance estimate s^2

b. The sample range divided by 4

c. The median absolute deviation

d. A default value of 1.0

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11 A sample of 100 lightbulbs has mean life $\bar{x} = 1200$ hours and unbiased sample variance $s^2 = 2500$. Find the 90% confidence interval for the true mean life.

a. $[1191.8, 1208.2]$

b. $[1180.0, 1220.0]$

c. $[1190.2, 1209.8]$

d. $[1195.0, 1205.0]$

12 What is the correct probabilistic interpretation of a 95% confidence interval $[L, U]$ for μ?

a. In repeated sampling, 95% of such constructed intervals will contain the true parameter μ

b. There is a 95% probability that μ lies in this specific interval

c. 95% of population values fall between L and U

d. The sample mean falls in $[L, U]$ 95% of the time

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